E-COMMERCE SALES ANALYSIS DASHBOARD

Comprehensive Business Case Study, Problem Statement & Functional Requirements

1. Executive Summary

In the competitive e-commerce landscape, data-driven decision-making is critical to optimizing operational efficiency, maximizing profitability, and identifying high-growth product categories and geographic markets. This business case study and problem statement document establishes the foundational framework for the **E-Commerce Sales Analysis Dashboard**. Designed to mirror executive-level intelligence needs, the dashboard consolidates multi-dimensional sales, profit, quantity, and order data into a unified, interactive analytical interface.

2. Problem Statement & Business Objectives

The Business Challenge: Leadership and regional managers face information silos and fragmented reporting across years, product hierarchies, and geographic territories. Without a centralized interactive interface, tracking Year-over-Year (YoY) growth, category profit margins, and state-level performance bottlenecks becomes time-consuming and prone to oversight.

Core Objectives:

- **Unified KPI Tracking:** Real-time visibility into Sales, Profit, Quantity, Total Orders, and Profit Margin alongside YoY growth metrics.
- **Temporal & Trend Analysis:** Monthly sales and profit trajectory breakdown to identify seasonal peaks and sales velocity.
- **Category & Portfolio Optimization:** Evaluate category-wise profit contribution and sales share to rebalance inventory and marketing investments.
- **Geographic & Granular Insights:** Analyze state-level revenue distribution and identify top 10 sub-categories driving profitability.
- **Dynamic Interactivity:** Implement cross-filtering capabilities across Years, Regions, and Customer Segments.

3. Dashboard Architecture & Component Specifications

The dashboard layout is structured into logical visual zones mirroring the reference interface:

Component Name	Visual Type	Functional Description & Metrics
KPI Cards Banner	Card / KPI Box	Displays aggregate Sales ($), Profit ($), Quantity (#), Orders (#), and Profit Margin (%) with YoY growth indicators.
Sales & Profit Analysis	Combo / Clustered Column Chart	Tracks monthly performance comparing Sum of Sales vs. Sum of Profit across the calendar year (Jan–Dec).
Category Wise Profit	Waterfall / Column Chart	Breaks down profit contributions across Technology, Office Supplies, and Furniture with Grand Total.
Category Wise Sales Share	Donut / Pie Chart	Displays percentage sales distribution across product categories (Furniture, Office Supplies, Technology).
Sales by State	Filled Map Chart	Geographic heatmap showing state-wise revenue distribution across the United States.

Top 10 Sub Categories	Horizontal Bar Chart	Ranks the top performing sub-categories (e.g., Chairs, Phones, Storage) by total sales volume.
Global Filters Panel	Interactive Slicers	Dynamic filters for Year (2011–2014), Region (Central, East, South, West), and Customer Segment.

4. Analytical Problem Scenarios & Exercises

To validate dashboard proficiency and test analytical acumen, data analysts and stakeholders are expected to solve the following structured problem scenarios derived from the dataset:

Problem 1: Seasonal Revenue & Profit Divergence

- *Scenario:* Using the monthly Sales & Profit Analysis combo chart, identify the months exhibiting the highest revenue volume versus those exhibiting margin compression or profit dips.
- *Task:* Calculate the profit-to-sales conversion ratio for November and determine whether high discount events correlate with reduced profit margins.

Problem 2: Category Profitability Paradox

- *Scenario:* The Category Wise Sales Share shows Furniture capturing ~32.4% of total sales, yet the Category Wise Profit chart reveals Furniture generates only a fraction of total profit compared to Technology and Office Supplies.
- *Task:* Formulate a strategic recommendation regarding product discounting and shipping cost allocation for Furniture sub-categories (e.g., Bookcases and Tables).

Problem 3: Geographic Concentration & Regional Expansion

- *Scenario:* Analyzing the Sales by State map alongside regional filters reveals stark disparities in market penetration between coastal states and central states.
- *Task:* Identify the top 3 states driving sales in the Western region and propose targeted logistical adjustments to improve fulfillment speed and reduce overhead.

Problem 4: Sub-Category Velocity Assessment

- *Scenario:* The Top 10 Sub Categories chart highlights Chairs and Phones as top revenue generators.
- *Task:* Using order quantity and discount data, rank the top 5 sub-categories by net profit margin and identify any loss-making sub-categories that should be deprioritized.

5. Expected Deliverables & Evaluation Criteria

Solutions submitted against this problem statement will be evaluated based on the following criteria:

1. **Data Accuracy:** Correct extraction and aggregation of metrics matching the interactive dashboard figures.
2. **Visual Interpretation:** Clear linkage between dashboard visual elements (Combo charts, Donut charts, Maps) and underlying business insights.
3. **Strategic Recommendations:** Actionable business proposals to improve profitability, optimize discounting strategies, and reallocate regional marketing spend.